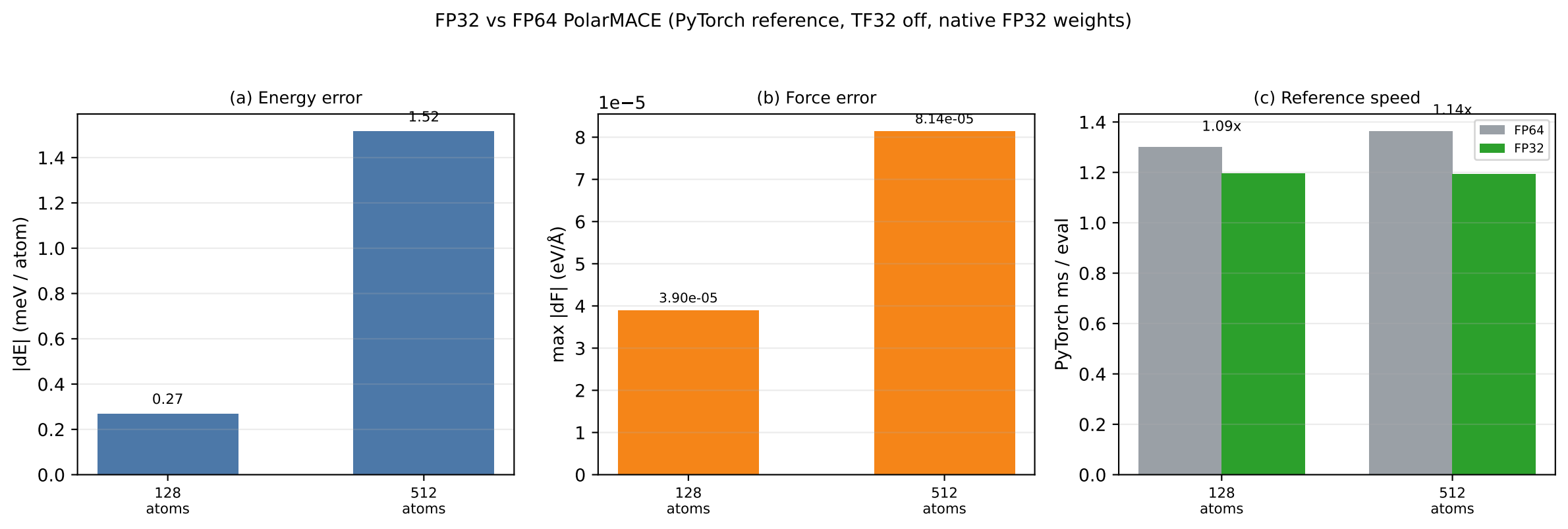
FP32 vs FP64 PolarMACE (PyTorch reference, TF32 off, native FP32 weights)
(a) Energy error
|dE| (meV / atom)
0.27
1.52
128 atoms
512 atoms
(b) Force error
1e−5
max |dF| (eV/Å)
3.90e-05
8.14e-05
128 atoms
512 atoms
(c) Reference speed
PyTorch ms / eval
1.09x
1.14x
FP64
FP32
128 atoms
512 atoms